

1 Physical Primitives of Mind: Modeling
2 Consciousness via Symmetries and Landauer
3 Erasure

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6 **Abstract**

7 We present a mathematical synthesis and formal Lean 4 verifi-
8 cation demonstrating how cognitive architectures can emerge as at-
9 tractor states from established physical primitives, without introduc-
10 ing new fundamental physics. Starting from the Poincaré group of
11 spacetime symmetries and finite phase space, we construct a frame-
12 work for predictive processing. Spontaneous symmetry breaking gen-
13 erates necessary topological boundaries, and finite phase space sub-
14 jects these boundaries to Landauer’s thermodynamic erasure. Agency
15 arises when systems evolve along paths of least action, minimizing en-
16 tropy production via internal structural resonance with external per-

17 turbations. Scaling this dynamic to macroscopic levels aligns with
18 emerging paradigms in neuroscience—specifically classical electrody-
19 namic fields (ephaptic coupling)—and Kuramoto phase synchroniza-
20 tion, culminating in a topological fixed point: a unified “Self.” The
21 formalization is reported with its scope attached: the development
22 declares no axioms and every structure carrying physical content is
23 inhabited by a witness, while the two links that remain assump-
24 tions—that the field reaches the synchronized configuration, and that
25 a global section of the probability sheaf is the unity of experience—are
26 marked as such throughout rather than absorbed into the deriva-
27 tion. As a corollary, we show why rigid von Neumann architectures
28 (e.g., GPUs) are thermodynamically disadvantaged or structurally
29 constrained from reaching these specific continuous thermodynamic
30 attractors.

31 1 Introduction

32 The pursuit of a physical theory of consciousness typically involves search-
33 ing for the biological correlates of mind or treating subjective experience as
34 an abstract computational algorithm. This paper asks a more fundamental
35 question: If we assume only basic physical primitives—continuous symme-
36 tries (the Poincaré group), geometry, and thermodynamics—is the emergence
37 of a unified, perceiving entity a natural physical attractor?

38 We propose a framework where the structural properties of consciousness

are derived as physical attractor states. Establishing a deductive chain from basic principles, we demonstrate how continuous symmetries break to form boundaries, how those boundaries dissipate free energy as non-equilibrium steady states, and how their evolution along paths of least action results in physical resonance with their environment. This eventually scales into the macroscopic electromagnetic unity proposed by emerging field theories of mind. To avoid critiques of “math theatre,” we delineate between literal physical mechanisms (e.g., thermodynamic erasure via Landauer’s principle, boundaries from spontaneous symmetry breaking) and theoretical mappings, utilizing Lean 4 formalization as our mathematical anchor. We adopt a perspective aligned with Russellian Monism: objective measurements of macroscopic electromagnetic fields constitute the third-person “outside,” while the system’s internal physical deformation to reach structural resonance constitutes the first-person intrinsic experience.

2 Formalization in Lean 4

A core contribution of this work is the verification of our deductive chain using the Lean 4 interactive theorem prover. This approach builds on recent efforts to formalize consciousness and brain semantics, such as the ‘OnlyOne’ system (Scherf, 2026) and sheaf-theoretic models of neural manifolds (Inoué, 2026). It is necessary to explicitly distinguish which claims have been mathematically derived as theorems within the proof assistant, and which remain

60 irreducible physical postulates. Table 1 maps the primary claims of this
 61 paper to their current formalization status.

Table 1: Mapping of physical claims to Lean 4 formalization status. The development declares **no axioms**: `#print axioms` on any result above reports only `propext`, `Classical.choice` and `Quot.sound`. Earlier drafts carried five axioms, three of them inconsistent; all five became class fields, so a postulate is an obligation on each instance rather than a global assertion (Section 2.1), and two of the five have since been eliminated entirely (Derivation 5). “Theorem” still does not mean “physically established”: the postulates live in hypotheses and instance obligations, and every such class is inhabited by a witness in `Examples.lean`, so no row is vacuous.

Conceptual Claim	Lean 4 Implementation	Status
Pointwise Vacuum Minimization	<code>pointwise_vacuum_of_global_min</code> (Phase1), potential energy defined as an integral; holds a.e., and a double-well witness proves the a.e. cannot be strengthened	Theorem
Landauer’s Principle	Derived from statistical mechanics (Phase3); the axiom-free reversibility core becomes a heat bound via the field <code>StatisticalMechanics.heat_eq</code>	Theorem + instance postulate
Symmetry Breaking implies Boundaries	Derived from pointwise potential minimization (Phase1); witnessed on a double well with a $\mathbb{Z}_2$ symmetry whose minimisation hypothesis is proved, not assumed	Theorem
Structural Resonance KL Bound	Gibbs’ inequality proved; the bound is the field <code>StructuralResonance.kl_bound</code> (Phase3_KLBound)	Theorem + instance postulate
Structural Resonance via Least Action	Gradient flow of σ in coupling space (Phase8), linked to <code>entropy_production_rate</code> ; finite substrates	Theorem
Phase-locking minimizes entropy production	For a <i>reciprocal</i> coupling kernel, the phase-locked field has the least entropy production of <i>any</i> field (<code>phase_locked_minimizes_entropy_of_symm</code> , Phase8 §2a). The mean-matching condition the earlier form assumed over the comparison class is derived, not imposed: the coupling term is anti-symmetric under exchanging its two sites, so total drift is conserved across phase fields (<code>total_drift_eq_of_symm</code>). Reciprocity is necessary — a one-way coupling on two sites strictly beats its own phase-locked state (Examples §16)	Theorem
Electrodynamic Discretization	Lebesgue-Bochner edge weights (Phase2) sitting on faces of the complex and partitioning the region’s stress-energy; mesh refinement converges, witnessed on $[0, 1)$. The scalar read off the stress-energy tensor is a marked modelling choice: the partition holds for any non-negative choice, and two instances differing only in it have different couplings (Examples §6)	Theorem + instance postulate

Conceptual Claim	Lean 4 Implementation	Status
Phase synchronization to Unity	Sheaf-theoretic gluing (Phase5) over a ThermodynamicCover , whose fields are now all local: the global section is constructed by probability_glue_unique and the former class field sync_to_section_eq is a theorem about it. Witnessed by two patches built from different mass profiles, each section constructed over its own patch, whose glued state is neither of them (Examples §14)	Theorem + instance postulate
Reflexive Topology (the Self)	Banach fixed point for a contracting self-prediction map (Phase6); metric, completeness and contraction are hypotheses, all three discharged on the three-site substrate (Examples §10), where global sections are proved to be exactly the finite measures on it, the metric is the uniform distance between their densities, and the self-prediction map contracts by exactly 1/2 without being constant. The avatar's link to the field is the predicate IsRestriction-Resonance , discharged on the same witness; under it a covering family of avatars determines the global state (eq_of_avatar_eq), and without it the Self theorem is unchanged (self_of_const-Resonance)	Theorem (witnessed)
Kuramoto Lyapunov descent	Descent chained to the minimum characterization by the rotating frame (Phase4_RotatingFrame); identical frequencies	Theorem
Kuramoto dynamics run	The initial value problem is well posed on all of $\mathbb{R}$: the field is globally Lipschitz and globally bounded (kuramotoField_lipschitz , kuramotoField_norm_le), and through every state at every instant passes exactly one trajectory, defined for all time (kuramoto_cauchy_problem). Along <i>every</i> trajectory of a zero-frequency system the dynamic potential converges (dynamic_potential_tendsto) and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is finite, equal to the potential's total drop (dissipation_integral_tendsto). The motion stops: Barbalat's lemma is proved here, Mathlib lacking it, and applied, so $\sum_i \dot{\theta}_i^2 \rightarrow 0$ along every trajectory (velocity_sq_tendsto_zero). Where it stops is not proved, and a two-oscillator trajectory that stops at the global minimum is exhibited instead (Examples §15)	Theorem (partial)
Continuous vs. Rigid Topology	Fixed wiring support at matched resource (Phase7_Rigidity), deriving the suboptimality Phase 7 assumed; typed continuum/discrete separation	Theorem

Conceptual Claim	Lean 4 Implementation	Status
Critical Coupling Threshold ($K_c = 2D$)	Self-consistency equation (<code>Phase8.SelfConsistency</code>): at or below K_c , threshold included, only $r = 0$ solves it; above K_c <i>exactly one</i> $r > 0$ does (<code>supercritical.fixed_point.existsUnique</code>), so the non-negative solutions are $\{0\}$ below and $\{0, r\}$ above; that r tends to 0 as $K \downarrow K_c$ and increases strictly with K (<code>coherent.branch.strictMono</code>); a solution is a density reproducing its own order parameter. Conditional on the von Mises stationary density, which is an input, and on no link to a trajectory	Theorem (conditional)

62 2.1 Soundness of the formalization

63 Formalization is only as trustworthy as the axioms it rests on, and an axiom
64 that is inconsistent silently makes every theorem downstream of it vacuous
65 while leaving the build green. We report a case of exactly this, found by
66 auditing our own development, because the failure mode is easy to reproduce
67 and we have not seen it discussed in the applied-formalization literature.

68 Earlier drafts of this work declared five axioms. Three of them were
69 individually inconsistent: each, on its own, proves **False**. All three shared
70 one root cause — an axiom that constrains a symbol it does not itself bind.
71 Concretely:

- 72 • `spontaneous_symmetry_breaking_pointwise_min` took the potential-
73 energy functional as an *implicit* argument occurring only in its hypoth-
74 esis `PotentialEnergy phi = V v0`. Instantiating that functional with
75 the constant map $\lambda_. V(v_0)$ discharges the hypothesis by `rfl` and yields
76 the conclusion for an arbitrary field. With $V(v) = v^2$, $v_0 = 0$ and $\phi \equiv 1$
77 this proves $1 = 0$.

- 78 • `landauer_heat_eq` equated the free class field `heat_dissipation` with
79 a bath-entropy difference *for every instance* of the class. An instance
80 with zero heat dissipation and a bath that gains one bit gives $0 = \log 2$.
- 81 • `kl_bound_axiom` asserted $\sigma \geq D_{KL}(P\|Q)/\Delta t$ for *all* P and Q , while σ
82 is a single real fixed by a class field. Since D_{KL} is unbounded above, a
83 point mass against a uniform distribution refutes it.

84 The remaining two axioms, `phase_invariant_periodic` and `sync_to_section_eq`,
85 had the same shape — global assertions pinning free fields of `LocalSectionSynchronization`
86 — and are refutable as soon as the probability presheaf admits two distinct
87 global sections. (Both were demoted to class fields at the time; both have
88 since been removed from the class altogether, for a separate reason given in
89 Derivation 5.)

90 The correction is uniform, and we state it as a rule we now follow: *a*
91 *physical postulate that mentions a class field must be a field of that class,*
92 *never a standalone axiom quantified over all instances.* A postulate should
93 be an obligation each model discharges, not a global claim about every model.
94 Recast this way, the development declares no axioms at all. Two of the five
95 became genuine theorems — pointwise vacuum minimization is now proved
96 from $\int f = 0 \wedge f \geq 0 \Rightarrow f = 0$ a.e., and Gibbs' inequality was already one
97 — and the other three became instance obligations.

98 This makes non-vacuity the live question, since an uninhabitable class
99 makes its theorems vacuous just as surely as an inconsistent axiom did.

100 `Examples.lean` therefore constructs witnesses: a one-bit erasure system with
 101 a genuine bath that discharges `heat_eq` and reproduces $\Delta Q = k_B T \ln 2$;
 102 a perfectly resonant system discharging `kl_bound`; and a scalar field on
 103 a one-point spacetime discharging `ActionPrinciples`. Derivation 5's glu-
 104 ing structures are witnessed too, which was until recently the largest open
 105 gap: `Examples.lean` builds a `ThermodynamicCover` over a three-site sub-
 106 strate carrying the discrete topology and its Borel σ -algebra, covered by two
 107 *overlapping* open patches whose intersection is a single shared site. The
 108 local sections are germ families of a phase-indexed finite measure $\mu_\theta =$
 109 $(1 + \cos \theta)^+ \delta_{\text{mid}}$ pushed into the sheafification of the probability presheaf; the
 110 family is provably non-constant in θ (`phaseMeasure_not_const`), so the pe-
 111 riodicity obligation of the constructor `LocalSectionSynchronization.of-`
 112 `InvariantMeasure` is discharged by an argument about `cos` rather than satis-
 113 fied trivially, and `thermodynamic_equilibrium` is *derived* from `phase_locked minimizes_potential`
 114 rather than assumed. `global_section_from_thermodynamics` then fires on
 115 this instance and glues the two patch-local sections into a unique global one.
 116 Two further covers answer the follow-up question of how non-degenerate such
 117 a witness can be made. `Examples.lean` §13 has a negative half worth record-
 118 ing: the phase field can be made non-constant as a function, but not as a
 119 physical state, because `thermodynamic_equilibrium` forces every instance
 120 to a phase-locked configuration. `Examples.lean` §14 is built the other way
 121 round, from two patches with independently chosen mass profiles that meet
 122 only on the site they share, and the section the gluing returns is neither of

123 them. That is the same kind of result as `contracting_implies_const` in
124 Derivation 6 — a proof that the witness is as strong as the class permits,
125 rather than a witness that could be improved by trying harder.

126 What the witnesses establish is that the classes are inhabitable, so the
127 theorems about them are not vacuous. What they do not establish is that
128 a cortex is such an instance. In the cover above the phase field is already
129 constant — every patch locked at $\theta = 0$ — which is precisely what makes
130 the equilibrium obligation dischargeable; getting a physical system into that
131 configuration is the content of the informal dynamical argument, not of the
132 Lean development.

133 The same audit turned up a defect of a different kind, worth record-
134 ing because it is not an axiom problem and would not have been caught
135 by `#print axioms. mesh_refinement_convergence` — the claim that dis-
136 crete edge weights converge to the continuous energy as the mesh is refined
137 — sat in a “Conjectures” section as an unproved `def ...: Prop`. It was
138 not merely unproved but false. Nothing required the edge regions to cover
139 the manifold, and $\text{diam } \emptyset = 0$, so the triangulation whose edge regions are
140 all empty satisfied the mesh hypothesis while contributing nothing to the
141 sum, forcing $|0 - \int f| < \varepsilon$ for every $\varepsilon > 0$. Its binder was dead as well: it
142 quantified over a triangulation and then resolved the vertex type by instance
143 search rather than through the bound variable, so the variable was never
144 used. `Phase2_MeshConvergence.lean` replaces it with a theorem. Conver-
145 gence is now stated along a *sequence* of triangulations of a named region

¹⁴⁶ S , each required to be regular — edge regions measurable, without self-
¹⁴⁷ loops, disjoint across distinct edges, covering S , and anchored to the sim-
¹⁴⁸ plicial data in the sense that a vertex lies in the closure of every region
¹⁴⁹ of an edge at it. Closure rather than membership is forced: cells are dis-
¹⁵⁰ joint, so a shared vertex belongs to at most one of them. The manuscript’s
¹⁵¹ discrete energy $\frac{1}{2} \sum_u \sum_v w(u, v) f(u)$, which samples only the first endpoint
¹⁵² and is therefore not symmetric, is shown to equal exactly the midpoint rule
¹⁵³ $\sum_{u < v} w(u, v) \frac{1}{2} (f(u) + f(v))$ over unordered edges; without that identifica-
¹⁵⁴ tion, reading the double sum as a Riemann sum would be a category error.
¹⁵⁵ At fixed mesh the error is at most $\varepsilon \mu(S)$ with ε a modulus of continuity of
¹⁵⁶ the integrand, and convergence follows for uniformly continuous integrands
¹⁵⁷ as the mesh shrinks. The regularity conditions pull against each other, so
¹⁵⁸ **Examples.lean** §6 discharges them with the uniform partition of $[0, 1)$ into
¹⁵⁹ half-open cells and checks that the approximation genuinely moves — $\frac{1}{2}$ at
¹⁶⁰ $N = 1$ and $\frac{1}{4}$ at $N = 2$ for the tent function $x \mapsto |x - \frac{1}{2}|$ — rather than
¹⁶¹ converging trivially. What is proved is convergence, not the $O(1/N^2)$ rate
¹⁶² the accompanying simulation measures; that would need a midpoint error
¹⁶³ term with a second derivative, which we have not developed.

¹⁶⁴ A third defect was one of bookkeeping across files, and it is the kind that
¹⁶⁵ survives any number of green builds because each half is individually cor-
¹⁶⁶ rect. The development carried *two* Kuramoto potentials. Lyapunov descent
¹⁶⁷ along the dynamics, $\dot{V} = -\sum_i \dot{\theta}_i^2$, was proved for the full potential $V(\theta) =$
¹⁶⁸ $-\frac{1}{2} \sum_{i,j} A_{ij} \cos(\theta_i - \theta_j) - \sum_i \omega_i \theta_i$, while the characterization of the minimum

169 — phase-locked in both directions — and everything downstream of it, in-
 170 cluding Derivation 5’s gluing, concerned the coupling term alone. Nothing
 171 chained them, and the gap was not cosmetic: `kuramoto_potential_unbounded_below`
 172 now proves that as soon as one $\omega_i \neq 0$ the full potential is unbounded below
 173 — walk θ out along ω and the bounded cosine sum cannot compensate — so
 174 it has no minimum for phase-locking to attain, and the sentence “the phase-
 175 locked state minimizes the Lyapunov potential” was, read literally, false of
 176 the functional the descent theorem was about. `Phase4_RotatingFrame.lean`
 177 closes this by formalizing the standard reduction $\theta_i \mapsto \theta_i - \Omega t$. For identical
 178 natural frequencies $\omega \equiv \Omega$, a trajectory maps to a trajectory of the zero-
 179 frequency system — which is exactly the system Derivation 5 already worked
 180 with — the two potentials coincide there, and the descent result transports
 181 onto the coupling potential whose minimizers are the phase-locked states.
 182 Phase differences, phase-locking and the magnitude of the order parameter
 183 are all proved invariant under the change of frame, so nothing observable
 184 is lost in the reduction. `Examples.lean` §7 runs the chain end to end on
 185 two oscillators with a common frequency, discharging every hypothesis. The
 186 restriction is real and we state it plainly: the reduction is exact only for
 187 identical frequencies. A genuine spread of natural frequencies leaves residual
 188 detunings in the reduced system, phase-locking then requires coupling above
 189 a critical K_c , and K_c is nowhere in our Lean development — it is studied
 190 numerically, as Table 1 records.

191 A fourth defect was dimensional, and it concerned the one row of Table 1

192 that is still numerical rather than proved. `Phase8_ContinuousField.lean`
 193 defines the critical coupling $K_c = 2D$ and a predicate for being above it.
 194 That predicate compared $\int_x \int_y K(x, y)$ against $2D$ — but a double integral
 195 against an unnormalized measure carries the substrate’s mass twice over,
 196 so a constant kernel K on a substrate of total mass m contributed Km^2 .
 197 The comparison therefore depended on how large the substrate was rather
 198 than on how strongly it was coupled, and any kernel whatever could be
 199 pushed above threshold by inflating m . `Examples.lean` §9 exhibits the fail-
 200 ure concretely: on a two-site substrate carrying counting measure, a sys-
 201 tem coupled at $K = 1/2$ against a threshold of $K_c = 1$ — a factor of two
 202 *below* threshold — had an unnormalized double integral of $2 > 1$ and so
 203 satisfied the old predicate. The fix is to require the substrate’s measure to
 204 be a probability measure, which is the normalization in which the mean-
 205 field derivation of $K_c = 2D$ is carried out; `mean_field_coupling_const`
 206 then gives that a constant kernel has mean-field strength exactly K , and
 207 `exhibits_phase_transition_const_iff` reduces the predicate to $K > 2D$.
 208 We flag what this does and does not do. It makes the predicate say something
 209 whose truth depends only on the physics, and §9 witnesses it on both sides of
 210 the threshold. It does not by itself prove $K_c = 2D$: the content of that value
 211 is the bifurcation of the self-consistency equation $r = I_1(Kr/D)/I_0(Kr/D)$
 212 at $K = 2D$, which is a separate question from whether the predicate is di-
 213 mensionally sound. That bifurcation is proved below, in both directions,
 214 from an assumed stationary density; deriving the density itself needs the

215 Fokker–Planck operator and is not done.

216 We draw no grand moral from this beyond the obvious one. The value
217 of a proof assistant here was not that it certified our argument, but that it
218 made a specific, checkable claim about what had been assumed — and that
219 claim turned out to be false in a way no amount of reading the prose would
220 have revealed.

221 3 Axiom 1: Symmetries and Phase Space

222 Our starting premise relies on fundamental geometry. In our Lean formaliza-
223 tion (`Phase1Primitives.lean`), continuous spacetime symmetries (e.g., the
224 Poincaré group) act upon field states, and the invariance of the system’s total
225 energy under that action is recorded by the class `SymmetryInvariantAction`.
226 Physically, it is such an invariance that Noether’s theorem converts into the
227 conservation of energy and momentum. The scope of the *formal* claim is nar-
228 rower and we state it plainly: the invariance is a class field, i.e. an assumption
229 a model must discharge, and no conserved quantity is ever constructed from
230 it. There is no Noether theorem in our Lean, and nothing downstream of this
231 section depends on one. The class is now inhabited — `Examples.lean` §11
232 exhibits a double-well potential $V(v) = (v^2 - 1)^2$ on a two-point spacetime
233 with the $\mathbb{Z}_2$ action $v \mapsto \pm v$, discharges the invariance because $(\pm v)^2 = v^2$,
234 and proves the vacuum manifold $\{-1, 1\}$ degenerate with the two minima ex-
235 changed by the symmetry, so an energy-minimising field and its image have

236 equal energy and differ. That is symmetry breaking exhibited rather than
 237 assumed; it is still not a conserved quantity, and $\mathbb{Z}_2$ being discrete there is
 238 no one-parameter family to differentiate along in the first place. Simultane-
 239 ously, discrete permutation symmetries dictate the statistical mechanics of
 240 any physical system, defining a strictly finite volume of phase space.¹

241 Because phase space is finite, any localized physical system possesses a
 242 mathematical limit on the amount of information (distinguishable states) it
 243 can embody at any given moment.

244 4 Derivation 1: Symmetry Breaking and the 245 Inevitability of Boundaries

246 A perfectly symmetric universe is the uniform vacuum; no distinct entities
 247 exist. For any dynamic process to occur, the continuous symmetries of the
 248 background field must undergo spontaneous symmetry breaking.

249 When a field drops to a lower, asymmetric energy state, topology dic-
 250 tates the natural creation of defects (such as domain walls or monopoles).

¹While the Poincaré group and discrete permutations serve as accessible physical examples, the underlying Lean formalization does not hardcode specific physics. It defines the system at a more primitive mathematical level, relying entirely on generic Pseudo-Riemannian manifolds (for spacetime) and finite measure theory (for statistical mechanics). What the formalization supplies is the vocabulary in which symmetries, conserved quantities and phase-space constraints can be *stated*; it does not derive them. All three of the classes involved are now inhabited in `Examples.lean` — the $\mathbb{Z}_2$ action and its invariance in §11, the real line with the Euclidean metric in §12 — but the instances are witnesses of inhabitability, not of physics: no theorem in the development consumes a symmetry group or a metric, so this section still contributes vocabulary rather than content.

251 This constraint necessitates the creation of a physical boundary—a localized
252 topological distinction between an “inside” and an “outside.” While such
253 a defect (e.g., a crack in a crystal lattice) is necessary to define a separate
254 entity, it is not sufficient for agency. The boundary is relational, defined by
255 the twisted geometry of the field surrounding it, and must be coupled with
256 non-equilibrium thermodynamics to exhibit active behaviors.

257 **5 Derivation 2: The Thermodynamic Con-** 258 **straint**

259 This newly formed topological boundary is subjected to gauge perturbations
260 from the “outside” environment. Because the internal phase space is finite
261 (Axiom 1), it cannot absorb an infinite sequence of incoming perturbations
262 and must overwrite old internal states.

263 By Landauer’s Principle, the erasure of physical information irreversibly
264 decreases internal entropy, mandating that the system exhaust heat into
265 the environment. Philosophers of physics (such as John Norton (Norton,
266 2005) and Orly Shenker (Shenker, 2000)) have critiqued Landauer’s princi-
267 ple, arguing that physicists conflate *logical* erasure (a many-to-one mapping
268 in abstract computational states) with *thermodynamic* erasure (a many-to-
269 one mapping in physical phase space). They argue that logical erasure could
270 conceptually be implemented by an injective physical process without dissi-
271 pating heat.

272 Our Lean 4 formalization (`Phase3.CombinatorialThermodynamics.lean`)
 273 addresses this critique by defining erasure as a non-injective mapping on the
 274 *physical* state space ($\neg \text{Function.Injective } t$). We formalize the global evolu-
 275 tion of the bipartite universe (System + Bath) as strictly injective, repre-
 276 senting Liouville’s theorem. Lean verifies that heat dissipation is not tied
 277 to abstract logical erasure, but is a mandatory mathematical consequence of
 278 any local physical state space contraction under globally reversible dynamics.
 279 If a cognitive system’s internal topology forces a contraction of its physical
 280 state space to accommodate structural resonance, heat dissipation becomes a
 281 thermodynamic requirement. A stable boundary thus constitutes a dissipative
 282 structure (Prigogine & Stengers, 1984), operating as a non-equilibrium
 283 steady state that consumes free energy to pay the thermodynamic cost of
 284 erasure. Continuous thermal fluctuations play a constructive role in estab-
 285 lishing boundaries between states while contributing to the system’s flexible,
 286 adaptive response.

287 **6 Derivation 3: Structural Resonance as a** 288 **Primitive**

289 In physical systems, time evolution follows a deterministic trajectory that
 290 minimizes thermodynamic entropy production. To reach this minimal dissi-
 291 pation state, the internal geometry of the structure must physically deform.

292 If we model the external environment as a stochastic driving force $P(ext)$

293 and the system’s internal state transitions as $Q(int)$, entropy production σ is
 294 strictly bounded from below by the Kullback-Leibler divergence $D_{KL}(P||Q)$.
 295 Gradient descent on σ necessarily drives $D_{KL}(P||Q) \rightarrow 0$, physically forcing
 296 the internal transition probabilities to match the statistical symmetries of
 297 incoming perturbations. We define this physical phenomenon as *structural*
 298 *resonance*.

299 This system does not possess a teleological drive to “survive”; rather,
 300 structurally mirroring the outside world is simply the lowest-energy attrac-
 301 tor state. This provides a thermodynamic mechanism for both the General
 302 Resonance Theory of Consciousness (Hunt & Schooler, 2019) and the Free
 303 Energy Principle (Friston, 2010; Friston & Kiebel, 2009; Hohwy, 2013; Clark,
 304 2016; Spisák & Friston, 2026). What neuroscience observes as “prediction er-
 305 ror minimization” is a physical system undergoing gradient descent to avoid
 306 dissipating heat against an unpredictable environment.

307 This necessity also addresses the symbol grounding problem (Harnad,
 308 1990). Internal states are physically indexed to the environment by the ener-
 309 getic imperative to avoid heat dissipation, rather than through arbitrary syn-
 310 tactic assignment. Manipulating abstract variables or counterfactuals forces
 311 the system to maintain internal states that deviate from immediate sensory
 312 resonance, explaining the continuous injections of free energy required for
 313 abstract symbolic reasoning.

314 7 Derivation 4: Macroscopic Scaling and Elec- 315 trodynamics

316 To resonate with highly complex environments, the system must scale up,
317 coupling millions of minimal dissipative structures. A highly coupled network
318 cannot allow all of its nodes to reach their lowest energy states simultane-
319 ously. This creates *geometric frustration*, resulting in a jagged phase space
320 of near-equal energy states that provides the physical capacity for complex
321 memory.

322 To manage this frustrated phase space without excessive dissipation, the
323 system undergoes coarse-graining (governed by the Renormalization Group).
324 At the macroscopic scale of biology, emerging paradigms in neuroscience sug-
325 gest these topological constraints manifest as classical continuous electromag-
326 netic fields. Building on seminal EM field theories of consciousness (McFad-
327 den, 2002; Pockett, 2012) and recent work demonstrating that macroscopic
328 electric fields (ephaptic coupling) govern representational drift, form memory
329 networks, and sculpt neural infrastructure (Pinotsis & Miller, 2022, 2023a,b),
330 we posit that *the continuous macroscopic EM field is the primary dissipative*
331 *structure interacting with the universe*. Instead of discrete wires, neurons in-
332 duce fields that shift the membrane potentials of neighboring cells. Modeling
333 and *in vitro* experiments confirm these endogenous fields are strong enough
334 to guide network activity and contribute to brain complexity (Goldwyn &
335 Rinzel, 2016; Ruffini & Santarnecchi, 2020; Cunha & Lima, 2024; Fröhlich &

336 McCormick, 2010). As proposed by the CEMI field theory, the continuous
337 electromagnetic field enables spatial integration of information, providing a
338 physical medium for a unified conscious gestalt (McFadden, 2020).

339 This continuous medium is physically supported by astrocytic syncytia—
340 non-neuronal networks joined by gap junctions that provide macroscopic cy-
341 toplasmic continuity for energy and ion flows (Verkhratsky & Nedergaard,
342 2018). Managing the thermodynamic cost of this state requires sophisticated
343 micro-bioenergetics. Dynamic mitochondrial networks function as deep-level
344 Mitochondrial Information Processing Systems (MIPS), transducing biolog-
345 ical and environmental information to sustain the coherence of the macro-
346 scopic field (Picard & Shirihi, 2022). Discrete synaptic firings, coupled
347 with mitochondrial micro-mechanics, generate and sustain the field. Because
348 the external environment is continuous, achieving maximal thermodynamic
349 structural resonance is optimally realized within this continuous macroscopic
350 medium (van Bree et al., 2025).

351 **8 Derivation 5: Global Sections and the Unity**

352 **of Consciousness**

353 The system is now characterized by a continuous macroscopic electrodynamic
354 field generated by millions of frustrated micro-components, forced to resonate
355 with the environment.

356 The Kuramoto model (Kuramoto, 1984) of coupled non-linear oscilla-

357 tors guarantees that, past a critical coupling threshold, these distributed
 358 local fields undergo a phase transition into unified synchronization (Bauer et
 359 al., 2022). Our Lean proofs (`Phase4_KuramotoDynamics.lean`) confirm this
 360 phase-locked state minimizes the Lyapunov potential of the system, in the
 361 precise sense discussed in Section 2.1: the functional minimized is the cou-
 362 pling potential read in the frame rotating with the natural frequency, which
 363 `Phase4_RotatingFrame.lean` shows is the honest reduced description of a
 364 system of identical oscillators.

365 This phase transition mathematically glues local synchronized states into
 366 a singular *Global Section* of a probability sheaf (`Phase5_GlobalSection.lean`).
 367 The EM field acts as a single, indivisible topological space, providing the
 368 *Unity of Consciousness*—a necessary condition for integrated experience that
 369 solves the binding problem. While Global Workspace Theory (Baars, 1997;
 370 Dehaene & Marti, 2014) models this unity as a broadcast architecture and
 371 Integrated Information Theory (Tononi & Koch, 2016) quantifies it via Φ ,
 372 our derivation provides a continuous physical mechanism. The contrast is
 373 worth drawing carefully, because both theories are addressing the same ex-
 374 planandum from different directions.

375 We state the scope of the Lean result precisely, because it is narrower
 376 than the prose above. `ThermodynamicCover` demands that the cover’s phase
 377 field minimize the reduced Kuramoto potential, and `potential_min_iff_`
 378 `phase_locked` characterizes those minimizers exactly: they are the phase-
 379 locked configurations, in both directions. Every instance is therefore already

380 phase-locked (`ThermodynamicCover.phase_locked`), and the class's single
 381 condition on its local data — that patches at a common phase agree where
 382 they overlap (`section_agrees_of_phase_eq`) — makes the family of local
 383 sections compatible. `probability_glue_unique`, which is the sheaf property
 384 of the probability presheaf and contains no physics at all, then glues that
 385 family into exactly one global section.

386 An earlier version of this structure proved less than the prose claimed,
 387 and we record the correction because it changed what the theorem says
 388 rather than how it is stated. `LocalSectionSynchronization` used to carry
 389 a *global* section as a class field, together with a second field declaring every
 390 local section to be a restriction of it. The global object therefore existed
 391 before the sheaf condition was ever consulted, and `global_section_from_`
 392 `thermodynamics` could do no more than rule out competitors to an object
 393 the instance had supplied. Both fields are gone. Every field of the class
 394 now mentions a single patch or a single overlap; the global section is con-
 395 structed (`ThermodynamicCover.invariantMeasure`); and the removed field
 396 survives as a *theorem* about that construction (`ThermodynamicCover.sync_`
 397 `to_section_eq`), so the local sections are restrictions of one common measure
 398 because the sheaf condition says so, not because an instance was required to
 399 name one.

400 `Examples.lean` §14 is the witness that the difference is not notational.
 401 Two patches carry two different mass profiles, $(2, 1, 7)$ and $(5, 1, 3)$ across
 402 the three sites, agreeing only at the site they share, where both put mass 1

403 — which is what discharges the overlap condition, by computing two num-
 404 bers rather than by construction. The section the theorem returns has pro-
 405 file `(2, 1, 3)`: it is neither of the two profiles (`gluedW_ne_leftW`, `gluedW_ne_`
 406 `rightW`), and neither of them restricts correctly to both patches (`leftW_`
 407 `glues_nothing`, `rightW_glues_nothing`). The global state carries informa-
 408 tion that no single patch had. `Examples.lean` §13, built the older way
 409 through the constructor `LocalSectionSynchronization.ofInvariantMeasure`,
 410 is retained alongside it for the negative result it carries: its phase field is not
 411 the constant function — the second patch a full turn ahead of the first — and
 412 by `ThermodynamicCover.phase_locked` that is as far from constant as any
 413 instance can get, since no instance can carry phases differing by anything a
 414 measure is allowed to see.

415 Two hypotheses still carry physical content and both are instance obli-
 416 gations, stated as such. That synchronized patches agree on their overlaps
 417 is a modelling assumption, not a consequence of any dynamics we formal-
 418 ize. And `thermodynamic_equilibrium` assumes the cover already sits at
 419 the potential minimum; no theorem says an arbitrary system’s dynamics
 420 takes it there. “A thermodynamic phase transition into unity” is there-
 421 fore still doing causal work that no general Lean result supports, though
 422 it is no longer entirely without one: `Examples.lean` §15 exhibits a two-
 423 oscillator trajectory that starts a quarter turn out of phase and converges
 424 to the global minimum of the dynamic potential, with the order parameter
 425 tending to 1. That trajectory is not an isolated construction: `kuramoto_`

426 `cauchy_problem` proves the Kuramoto initial value problem well posed on
 427 all of $\mathbb{R}$, so through every initial state there passes exactly one trajectory,
 428 and the exhibited one is the only solution through its own. Nor is its set-
 429 tling down peculiar to it: along every trajectory of a zero-frequency system
 430 the dynamic potential converges, and the total dissipation $\int_0^\infty \sum_i \dot{\theta}_i^2$ is fi-
 431 nite and equal to the potential's total drop (`dynamic_potential_tendsto`,
 432 `dissipation_integral_tendsto`). The motion also stops: a finite integral
 433 does not by itself force its integrand to zero, but the dissipation rate is Lip-
 434 schitz in time, so Barbalat's lemma applies, and every trajectory satisfies
 435 $\sum_i \dot{\theta}_i^2 \rightarrow 0$ (`velocity_sq_tendsto_zero`; Mathlib has no Barbalat, so it is
 436 proved here as `tendsto_zero_of_lipschitz_of_integral_le`). What sepa-
 437 rates that from what `thermodynamic_equilibrium` assumes is one step, and
 438 we state it precisely: the trajectory comes to rest, but not necessarily at a
 439 *global* minimum of the potential — splay and twisted configurations are equi-
 440 libria too, and a trajectory resting at one converges to a value that is not the
 441 minimum. So no theorem of the form “every trajectory reaches the unified
 442 state” is available, or true. What Lean establishes about the gluing itself is
 443 the other half: *given* locally synchronized patches, a global state exists, is
 444 unique, and is produced by the gluing rather than presupposed by it.

445 Integrated Information Theory begins from phenomenological axioms and
 446 derives a scalar Φ measuring irreducibility of a system's cause-effect structure,
 447 with the exclusion postulate selecting a single maximum- Φ complex as the
 448 locus of experience. Our sheaf-theoretic global section plays a structurally

449 analogous role — it too picks out a distinguished unified object — but it
450 is arrived at from the opposite end: not postulated from the character of
451 experience, but forced by a thermodynamic phase transition in a physical
452 field. This buys two things IIT does not have. First, the unity condition
453 is checkable from field dynamics rather than requiring the computation of
454 Φ , which is intractable for systems of realistic size and whose definition has
455 undergone repeated revision (Tononi & Koch, 2016). Second, exclusion falls
456 out rather than being postulated: gluing either succeeds on a given cover or
457 it does not, so there is no separate principle needed to rule out overlapping
458 conscious subsystems. It also inherits IIT’s exposures. Critics have argued
459 that Φ can be high in systems with no plausible claim to experience, such
460 as large feedforward-free grids (Aaronson, 2014); our criterion is vulnerable
461 to the mirror objection, since a physical system could in principle satisfy
462 the gluing condition without being cognitively interesting. We take this
463 seriously rather than dismissing it: our reply is that gluing is necessary but
464 not sufficient, and that the topological boundary of Derivation 1 and the
465 reflexive structure of Derivation 6 are additional required conditions. IIT, by
466 contrast, treats Φ as both necessary and sufficient.

467 Global Workspace Theory faces a complementary difficulty. It specifies
468 a functional architecture — a limited-capacity workspace broadcasting to
469 specialized modules — without specifying what physical property of broad-
470 casting makes the broadcast content conscious, leaving the theory open to the
471 charge that it explains access consciousness while leaving phenomenal con-

472 sciousness untouched (Block, 1995). Our framework can be read as supplying
 473 the missing physical substrate for the workspace: the continuous macroscopic
 474 field *is* the medium in which integration is realized, and the $K_c = 2D$ thresh-
 475 old gives a physical condition for ignition rather than a functional stipulation
 476 of it. Where GWT is agnostic about implementation, we make a falsifiable
 477 commitment — unity requires a continuous physical medium, which predicts
 478 that a functionally equivalent discrete broadcast architecture would not be
 479 conscious. We regard this as a genuine cost of the position, not a free ad-
 480 vantage: it is the point at which our framework departs most sharply from
 481 computational functionalism and where it is most readily falsified. Contrast
 482 this derivation with panpsychism: consciousness is not assumed as a funda-
 483 mental property of matter. It requires the initial existence of a topological
 484 defect to establish a bounded “inside” (Derivation 1), followed by a thermo-
 485 dynamic phase transition into unity that occurs when integrated spatial cou-
 486 pling exceeds the thermodynamic noise threshold ($K_c = 2D$). Without both
 487 the foundational topological boundary and the subsequent mathematically
 488 defined phase transition, the system remains a fragmented, non-conscious
 489 collection of micro-states. A unified field of perception, however, is not yet
 490 a “Self.”

491 9 Derivation 6: Reflexive Topology and the 492 Emergence of the Self

493 While a global section provides Unity, a “Self” requires reflexivity. The
494 unified EM field is a dissipative structure that must resonate with external
495 perturbations while managing its own internal phase transitions. Time evo-
496 lution naturally drives the field to minimize thermodynamic friction with its
497 own internal state-changes.

498 This mandates *auto-resonance*: the field must predict itself. The “Self”
499 emerges as a higher-order topological feature when the internal geometry
500 folds to include a low-dimensional avatar of its own computational and causal
501 boundary within its predictive model (`Phase6_ReflexiveTopology.lean`)
502 (Levin, 2019). The system is unified, predictive, and explicitly maps its
503 own distinct existence from the environment. This architecture provides the
504 physical foundation for a self-aware consciousness, bounded by the continuous
505 measure thermodynamics of Landauer erasure (`Phase3_MeasureThermodynamics.lean`).
506 We are explicit about what carries the weight here. `reflexive_topology_implies_self`
507 is the Banach fixed-point theorem with the physical commitments in its hy-
508 potheses: the space of global sections is assumed nonempty, metric and com-
509 plete, and the predictive map is assumed to contract by $1/2$. `Examples.lean` §10
510 discharges all four on the three-site substrate of Derivation 5, and the metric
511 is no longer a placeholder. The section first pierces the sheafification: on
512 a discrete substrate every point has a smallest open neighbourhood, so the

mass a germ carries at a point is well defined, and `massEquivOn` proves that
 over *any* open the sections of the probability sheaf *are* the finite measures
 on that open, read as densities against counting measure — the global case
 being $U = \top$. Mathlib supplies no inverse to this sheafification — its `TopCat`
 sheafification has neither an adjunction nor an `isIso_toSheafify` — so the
 correspondence had to be constructed. The metric on sections is then the
 uniform distance between the densities of the measures they glue to — equiv-
 alent to their total variation on a finite substrate, though not equal to it —
 and it is proved complete by exhibiting the limit section of a Cauchy sequence
 of densities, not by the degenerate route of showing Cauchy sequences are
 eventually constant. The self-prediction map is the relaxation “average the
 current field with the baseline,” and we prove it contracts distances by *ex-*
actly one half, that it is not constant, and that its fixed point is unique. An
 earlier version of this witness used the 0/1 metric, under which every con-
 traction is constant; that theorem is kept in the source as the record of why
 the earlier witness was empty. A separate gap, closed at the same time, con-
 cerned the avatar rather than the fixed point. `ReflexiveBoundary` carries a
 field `auto_resonance` sending the global state to a section over the avatar re-
 gion, and nothing constrained it: a constant function was a legal avatar, and
`reflexive_topology_implies_self` consumes only the predictive model, so
 no theorem in the development mentioned the field the avatar is supposed to
 be reading. The constraint is now stated as a predicate rather than added
 to the structure — `IsRestrictionResonance`, that the avatar’s state is the

536 presheaf restriction of the global state — and two theorems say why it is
 537 the right one. In the positive direction, if a family of resonant avatars *cov-*
 538 *ers* the substrate then the global state is determined by their local readings
 539 alone (`eq_of_avatar_eq`): unity is not assumed alongside the avatars but
 540 recovered from them, by the same sheaf condition that Derivation 5 uses.
 541 In the negative direction, replacing any avatar by a constant leaves a legal
 542 `ReflexiveBoundary` with the same region, the same predictive model and
 543 the same Self (`self_of_constResonance`), so the fixed-point theorem cannot
 544 see the difference and the predicate is doing the work rather than restating
 545 it. Both halves are witnessed on the three-site substrate, where the avatar
 546 region provably separates two states of the field, so the condition is a con-
 547 straint there and not a formality. What is still not established is that any field
 548 dynamics produces this particular map or performs the restriction, or that
 549 the contraction constant $1/2$ is read off an entropy production rate rather
 550 than built into the map's definition. We also note a negative result found
 551 in the course of this work: the Lévy–Prokhorov metric, which the natural
 552 weak-convergence reading would suggest, does *not* support this argument on
 553 a discrete substrate, because the ε -thickening of a set is the set itself for $\varepsilon < 1$
 554 and the relaxation map is consequently not a Lévy–Prokhorov contraction.

555 10 Derivation 7: Continuous Stochastic Ther- 556 modynamics and Field Theory

557 To move beyond finite graph analogies, we model the macroscopic electro-
558 dynamic field as a *Continuous Neural Field*. True ephaptic coupling is a
559 continuous spatial phenomenon, not a discrete network. We extend the
560 discrete Kuramoto model to a continuous spatial coupling kernel $K(x, y)$
561 (`Phase8ContinuousField.lean`).

562 Subject to thermodynamic noise D , the system undergoes non-equilibrium
563 stochastic thermodynamics. Rather than a static Landauer bound, the field
564 dynamics minimize an entropy production functional σ :

$$\sigma = \int \frac{1}{D} \left(\omega(x) + \int K_t(x, y) \sin(\theta_y - \theta_x) dy \right)^2 dx \quad (1)$$

565 Biological fields dynamically deform their spatial topology $K_t(x, y)$ to un-
566 dergo gradient descent on this entropy production functional.

567 That gradient descent is a theorem rather than a modelling assertion,
568 for finite substrates. Reading σ as a functional of the coupling kernel at
569 fixed phases, `hasFDerivAt_sigmaOfKernel` computes its Fréchet derivative
570 in closed form,

$$\frac{\partial \sigma}{\partial K(a, b)} = \frac{2}{D} \left(\omega(a) + \sum_y K(a, y) \sin(\theta_y - \theta_a) \right) \sin(\theta_b - \theta_a), \quad (2)$$

571 and `sigmaOfKernel.eq.entropy_production_rate` identifies the functional
 572 being differentiated with `entropy_production_rate` itself whenever the sub-
 573 strate carries counting measure. `structural_resonance_decreases_entropy_production`
 574 then concludes that σ is non-increasing along any coupling trajectory relax-
 575 ing down its own gradient at a positive rate c . Earlier drafts proved this
 576 only for an abstract functional S on an abstract inner-product space with
 577 the gradient supplied as data; the identification of that S with σ was infor-
 578 mal, and Table 1 flagged the row accordingly. Two limitations stay explicit,
 579 and the first is narrower than we previously reported. The derivation is for
 580 *finite* substrates. We had recorded the obstacle to the continuum case as
 581 differentiation under the integral sign with respect to the kernel; that was
 582 a misdiagnosis. σ is not a general functional that happens to be written as
 583 an integral — it is a quadratic functional of an *affine* image of the kernel,
 584 since the drift is a fixed ω plus a linear operator applied to K , and σ is a
 585 constant times the squared norm of the drift. A squared norm is Fréchet
 586 differentiable on any real inner-product space and an affine map is its own
 587 derivative, so the composite differentiates by the chain rule with no limiting
 588 argument. `hasFDerivAt.quadratic_of_affine` states that in full generality,
 589 and `hasFDerivAt.sigmaOfKernel_of_operator` re-derives the finite gradient
 590 from it, showing that the closed form displayed above is a convenience rather
 591 than the content. What the continuum case still needs is therefore one object:
 592 the drift map $K \mapsto (x \mapsto \int K(x, y) \sin(\theta_y - \theta_x) dy)$ as a bounded operator
 593 $L^2(\mu \otimes \mu) \rightarrow L^2(\mu)$, which is a Cauchy–Schwarz estimate — the integral ker-

nel is bounded by 1, so on a finite measure space the operator norm is at most
 $\mu(M)^{1/2}$. We have not built it, because it requires constructing a continu-
 ous linear map between L^p spaces and we have not done that bookkeeping;
 we record the corrected scope because “construct one bounded operator”
 and “differentiate under the integral sign” are different sizes of task. The
 second limitation is unchanged: the phase field is held fixed throughout, so
 this is plasticity of the coupling at frozen phases, not joint (θ, K) dynamics.
Examples. §5 exercises the whole chain on a two-site substrate whose
 coupling relaxes as e^{-t} and whose entropy production decays as e^{-2t} , con-
 firming that the gradient is not identically zero and that the descent is strict
 rather than vacuously monotone.

A second result about σ concerns where the descent is headed, and it has
 been strengthened since the last draft in a way that also corrects an estimate
 we had recorded. That the phase-locked field minimizes σ was proved only
 against competitors sharing its spatial mean drift — a restriction that left
 the theorem close to Jensen’s inequality and no more — and we had judged
 that lifting it would require modelling how the mean drift varies with the
 competitor. That judgement was wrong. For a *reciprocal* kernel, one with
 $K(x, y) = K(y, x)$, the restriction is not a restriction: the coupling integrand
 $K(x, y) \sin(\theta_y - \theta_x)$ is antisymmetric under exchanging its two sites, because
 K is symmetric while $\sin$ is odd, so Fubini identifies its double integral with
 its own negative and the total coupling contribution vanishes. The total drift
 $\int(\omega + \text{coupling})$ is therefore $\int \omega$ whatever the phases are (`total_drift_eq_of_`

617 `symm`): coupling moves drift between sites but neither creates nor destroys it.
 618 Evaluating that conserved quantity at the phase-locked configuration, where
 619 the drift is constant, pins its value, and the mean-matching hypothesis follows
 620 for every competitor. `phase_locked_minimizes_entropy_of_symm` is the un-
 621 restricted statement. Reciprocity is the physically intended condition — an
 622 ephaptic field acts on both sites of a pair alike, and it is the same symmetry
 623 `ThermodynamicCover.A_symm` already imposes in Derivation 5 — but it is
 624 not a formality, and we prove that rather than assert it. `Examples.lean` §16
 625 exhibits a *one-way* coupling on two sites, in which a feels b and b does not
 626 feel a , whose phase-locked field is strictly beaten: displacing the second phase
 627 to $-\pi/2$ cancels the first site’s natural drift while adding none at the sec-
 628 ond, halving σ . So “phase-locking minimizes entropy production” is false for
 629 general kernels and true for reciprocal ones, and the hypothesis cannot be
 630 removed by working harder.

631 We define the critical coupling threshold $K_c = 2D$. The phase transition
 632 to the unified “Global Section” occurs when the mean-field spatial coupling
 633 exceeds this thermodynamic noise threshold, linking the emergence of Unity
 634 to physical temperature and noise.

635 Earlier drafts left $K_c = 2D$ as a bare stipulation validated only by
 636 simulation. `Phase8_SelfConsistency.lean` now proves the half of it that
 637 is ordinary real analysis. For the noisy mean-field Kuramoto model with
 638 identical frequencies, the stationary density at order parameter r is the
 639 von Mises density $\rho(\theta) \propto e^{(Kr/D)\cos\theta}$, and self-consistency requires $r =$

$R(K, r) := I_1(Kr/D)/I_0(Kr/D)$. Mathlib has no modified Bessel functions,
 so we formalize that ratio directly as a ratio of integrals over $[-\pi, \pi]$. In-
 tegration by parts with $u = \sin$, $v = -e^{a \cos \theta}$ gives an exact identity with
 no boundary term, $\int \cos \theta e^{a \cos \theta} d\theta = a \int \sin^2 \theta e^{a \cos \theta} d\theta$, so R is a times
 the mean of $\sin^2$ under the von Mises weight. Bounding that mean by 1
 would put the threshold at D ; the true threshold is $2D$, and the entire fac-
 tor of two is the sharper bound $\mathbb{E}_a[\sin^2 \theta] \leq 1/2$, equivalently $I_2(a) \geq 0$.
 We prove it elementarily, by folding $[-\pi, \pi]$ onto $[0, \pi/4]$ in two reflections
 $(\theta \mapsto \pi - \theta, \text{ then } \theta \mapsto \pi/2 - \theta)$, after which the integrand is a product
 of $\cos 2\theta \geq 0$ with $2 \cosh(a \cos \theta) - 2 \cosh(a \sin \theta) \geq 0$ — the second factor
 because $0 \leq \sin \theta \leq \cos \theta$ there and $\cosh$ increases in $|\cdot|$. The conclusion,
`subcritical_fixed_point_eq_zero'`, is that for $K < K_c$ the incoherent state
 $r = 0$ is the *only* non-negative solution of the self-consistency equation; $r = 0$
 is exhibited as a solution, so the uniqueness claim is not vacuous. Three fur-
 ther facts about the threshold have since been proved, and they narrow the
 gap between this row and a full “Theorem” to a single named statement.
 First, the fold argument gives a *strict* inequality for $a > 0$ (`vonMisesC2_`
`pos`, hence $R(a) < a/2$), which settles the threshold itself: at $K = K_c$ the
 self-consistency map is tangent to the diagonal at the origin but falls strictly
 below it thereafter, so $r = 0$ remains the only solution and the bifurcation
 happens strictly *after* K_c , not at it (`fixed_point_eq_zero_of_le_critical`).
 Second, factoring $R(a) = a E(a)$ shows that a non-zero r solves the equation
 exactly when $E(Kr/D) = D/K$ (`coherent_iff_sRatio_eq`), so the coher-

ent branch is a level set of the single function $E(a) = \mathbb{E}_a[\sin^2 \theta]$. Third,
 and using only $E < 1/2$ together with compactness, every coherent solution
 tends to zero as $K \downarrow K_c$ (`coherent_branch_continuous_at_threshold`): the
 branch emerges continuously from the incoherent state rather than appearing
 at a finite distance from it, which is what makes the transition supercritical
 in the technical sense. That statement quantifies over *every* solution, so it
 does not presuppose the uniqueness it does not have. The level-set form
 also says exactly what uniqueness is: the injectivity of E on $(0, \infty)$. That
 is now proved, in the stronger form that E is strictly decreasing on $[0, \infty)$
 (`vonMisesSRatio_strictAntiOn`), and by an argument that avoids the route
 we had expected to need. Differentiating the exponential family in its natu-
 ral parameter gives $E'(a) = -\text{Cov}_a(\cos^2 \theta, \cos \theta)$, which costs differentiation
 under the integral sign and leaves a covariance that is not sign-definite point-
 wise: symmetrising with an independent copy gives $\frac{1}{2}\mathbb{E}[(X - X')^2(X + X')]$
 with $X = \cos \theta$, an integrand changing sign with $X + X'$. Nothing in the
 proof differentiates anything. The target is kept as the two-point inequality
 $C_2(a)Z(b) < C_2(b)Z(a)$ for $a < b$, and two moves settle it. *Fold first*: the
 two reflections that proved $I_2 \geq 0$ carry both integrals onto $[0, \pi/2]$, replacing
 $e^{a \cos \theta}$ by $W(a, \theta) = 2 \cosh(a \cos \theta)$ and confining $\cos \theta$ to $[0, 1]$, after which
 $\cos 2x - \cos 2y$ and $W(b, x)W(a, y) - W(a, x)W(b, y)$ have the same sign, both
 being governed by $\cos x$ against $\cos y$. This is the symmetrisation $X \mapsto -X$
 that the covariance form needed the tilt to perform, done once at the start; its
 analytic content is one identity, $2 \cosh X \cosh Y = \cosh(X+Y) + \cosh(X-Y)$

686 with $\cosh$ increasing in $|\cdot|$. *Then cross*: instead of integrating that inequality
 687 over a square, let α be the mean of $\cos 2x$ under $W(a, \cdot)$ and $x_0 \in [0, \pi/2]$ a
 688 point where $\cos 2x_0 = \alpha$, supplied by the intermediate value theorem; eval-
 689 uating at $y = x_0$ and integrating in x alone gives the result, since the term
 690 carrying $W(b, x_0)$ vanishes by the choice of α . No product measure and no
 691 Fubini appear. Run with $\cos \theta$ in place of $\sin^2 \theta$ the same argument needs no
 692 fold — $R(a) = \mathbb{E}_a[\cos \theta]$ is the mean of the variable the family is tilted by —
 693 and gives **besselRatio_strictMono**: R is strictly increasing on all of $\mathbb{R}$, the
 694 monotonicity of the Bessel ratio our earlier drafts recorded as unproved.

695 The converse half is **supercritical_fixed_point_exists**: for $K > K_c$
 696 there is an r with $0 < r \leq 1$ solving $r = R(K, r)$. We had expected
 697 this to require a second-order expansion of the Bessel ratio, and it does
 698 not. Writing $R(a) = a E(a)$ with $E(a) = \mathbb{E}_a[\sin^2 \theta]$ — the same factoriza-
 699 tion the integration by parts already supplies — the only facts needed are
 700 that E is continuous and that $E(0) = 1/2$, the latter because the weight
 701 is constant at $a = 0$. Since $K > 2D$ makes D/K strictly less than $1/2$,
 702 continuity gives $E > D/K$ on a neighbourhood of the origin, so the map
 703 overshoots, $R(K, r) > r$, for all small $r > 0$; at the other end $R \leq 1$ al-
 704 ways, because R is a mean of $\cos \theta$, so the map undershoots at $r = 1$.
 705 The intermediate value theorem closes the gap. Continuity of E is Math-
 706 lib's continuity of a parametric interval integral with jointly continuous in-
 707 tegrand, so no dominated-convergence argument is written by hand, and
 708 the existence proof never uses monotonicity of R : the fixed point is pro-

709 duced by the intermediate value theorem, not by iterating the map. That
 710 R is in fact strictly increasing is proved below, and no threshold result de-
 711 pends on it. `critical_coupling_is_threshold` packages the two halves,
 712 and `exhibits_phase_transition_coherent` applies the supercritical half to
 713 any substrate satisfying the predicate of the previous paragraph — so ex-
 714 ceeding K_c now implies the existence of a coherent stationary order pa-
 715 rameter rather than merely comparing against a named number. With
 716 uniqueness available both are sharpened — `supercritical_fixed_point_`
 717 `existsUnique`, `supercritical_solution_set` (the non-negative solutions
 718 are exactly $\{0\}$ below threshold and $\{0, r\}$ above it) and `exhibits_phase_`
 719 `transition_unique_coherent` — and the two monotonicities together fix
 720 the shape of the branch: for $K_1 < K_2$, $E(a_i) = D/K_i$ with E decreasing
 721 puts the concentrations in the order $a_1 < a_2$, and $r_i = R(a_i)$ with R increas-
 722 ing carries that to $r_1 < r_2$ (`coherent_branch_strictMono`). The coherent
 723 order parameter grows strictly with the coupling, which with the continuity
 724 at threshold is the qualitative bifurcation diagram.

725 One more step makes the equation mean in Lean what it means on paper.
 726 As a definition, `selfConsistency K D r` is just $R(Kr/D)$ — a function of
 727 r with a suggestive name, and nothing in the source says it has anything
 728 to do with an order parameter. `Phase8_SelfConsistency §8` supplies the
 729 identification. `vonMisesDensity` is the normalized weight, proved to be a
 730 genuine probability density on $[-\pi, \pi]$; `circularOrderParameter` is the av-
 731 erage of $e^{i\theta}$ against a density, the continuum analogue of the finite-system

732 `order_parameter_complex` of Phase4; and that average is proved to be ex-
 733 actly $R(a)$ — real, because the density is even, so the mean direction van-
 734 ishes. `fixedPoint_iff_selfReproducing` then states the equivalence the
 735 name promises: r solves $r = R(K, r)$ if and only if the von Mises density
 736 it induces has order parameter r . Read through this, the existence theorem
 737 says that above K_c there is a stationary density whose own order parameter
 738 is positive, and the solution $r = 0$ is identified with the uniform density,
 739 whose order parameter is zero.

740 What remains not established, and we separate it carefully. The sta-
 741 tionary density is an *input*: deriving it from the Langevin dynamics $d\theta =$
 742 $(\omega + K \cdot \text{mean-field}) dt + \sqrt{2D} dW$ needs the Fokker–Planck operator, existence
 743 and uniqueness of its stationary solution, and a spectral stability argument,
 744 none of which Mathlib has. Of the coherent branch we prove existence,
 745 uniqueness, continuous emergence at the threshold and strict growth in K ,
 746 but not that it is the dynamically selected one: nothing in that file men-
 747 tions a trajectory, so stability and selection are not statements it can make.
 748 And while `selfConsistency` is now connected to the order parameter of
 749 a *density*, it is connected to no *trajectory*: `circularOrderParameter` is an
 750 integral, `order_parameter_complex` is an average over finitely many oscilla-
 751 tors, and relating them is the mean-field limit — propagation of chaos for
 752 the finite Kuramoto system — which is a research programme rather than
 753 a lemma. That single gap is now the whole of what separates our threshold
 754 predicate from a statement about dynamics. Table 1 records the row ac-

755 cordingly as a theorem about the self-consistency equation from an assumed
756 density, not as a derivation of the threshold from the dynamics. Perceptual
757 anomalies such as visual snow can be modeled as boundary conditions where
758 microscopic thermal noise (D) fails to be fully coarse-grained away by the
759 continuous field, leaking across the K_c threshold into the macroscopic gestalt.

760 Clinical evidence for this phase transition is observed during general anes-
761 thesia. Work by Miller and colleagues (Bastos & Miller, 2021) demonstrates
762 that anesthetics like propofol induce unconsciousness not by suppressing neu-
763 ral firing, but by disrupting the phase alignment of high-frequency cortical
764 rhythms. Destroying the dynamic stability of these rhythms breaks the struc-
765 tural resonance of the continuous field, forcing the integrated spatial coupling
766 below the critical threshold K_c and fracturing the unified Global Section back
767 into disjoint, unconscious local fields.

768 The daily loss of consciousness during sleep can similarly be modeled as
769 an active physical disruption of the field. During sleep, astrocytic shrink-
770 age expands the extracellular space, allowing an influx of cerebrospinal fluid
771 (CSF) to flood the brain for metabolic waste clearance (Xie & Nedergaard,
772 2013). This influx physically alters the spatial medium, disrupting the contin-
773 uous structural resonance and preventing the coherent sheaf-theoretic gluing
774 required for conscious unity.

775 This mechanical framework models the temporal mismatch observed dur-
776 ing sudden awakenings (sleep inertia). When jolted awake, the locus coeruleus
777 floods the cortex with noradrenaline. High-frequency electrical rhythms

778 attempt to phase-align in milliseconds (Bastos & Miller, 2021), but the
 779 noradrenaline-driven astrocytic expansion required to physically squeeze the
 780 excess CSF out of the interstitial pathways operates on a much slower timescale
 781 of seconds to minutes (Xie & Nedergaard, 2013). During this lag, electrical
 782 oscillators attempt to synchronize across a waterlogged topology, resulting
 783 in the fragmented structural resonance experienced as grogginess. The uni-
 784 fied Global Section fully re-establishes only when the physical hardware (as-
 785 trocytic volume) finishes constricting to match the state of the electrical
 786 rhythms.

787 **11 Corollary: Hardware Thermodynamics and** 788 **Topological Rigidity**

789 Both biological tissue and artificial silicon hardware (like GPUs) are physical
 790 dissipative structures. When a GPU computes, its charge carriers undergo
 791 physical state transitions; by Landauer’s principle, this erasure irreversibly
 792 exhausts heat into the environment.

793 There is a fundamental thermodynamic distinction in the degrees of free-
 794 dom available to minimize this dissipation. In standard von Neumann ar-
 795 chitectures, the spatial topology (the silicon routing lattice) is rigidly fixed.
 796 The coupling matrix A_{ij} is constrained by a discrete layout that cannot be
 797 physically altered during operation. The brain’s own architecture provides
 798 empirical validation for this constraint. The cerebellum contains approxi-

799 mately 80% of the brain’s neurons arranged in a highly efficient, discrete,
800 feedforward crystalline circuit (Herculano-Houzel, 2009; Lange, 1975). Clin-
801 ical observations of cerebellar agenesis report largely preserved conscious
802 experience despite motor and cognitive deficits (Yu et al., 2015), and this
803 dissociation has been read as evidence for the importance of recurrent ar-
804 chitecture by IIT (Tononi & Koch, 2015) — an argument we borrow at the
805 level of the anatomical observation while declining its Φ -based explanation.
806 Lacking the dynamic, continuous, and recursive spatial coupling of the cere-
807 brum, it operates much like a biological GPU—highly optimized for discrete
808 processing, yet apparently not required for conscious experience. We take the
809 dissociation as evidence, not as confirmation: it is consistent with subjective
810 experience being a thermodynamic attractor of continuous topologies rather
811 than a byproduct of information processing, and it is the kind of observation
812 that would be hard to reconcile with the converse. It is a single anatomical
813 case read against a theory, and it bears the weight such a case can bear.

814 This constraint does not preclude artificial consciousness, but rather re-
815 stricts its hardware. To achieve the phase transition into a unified Global
816 Section, artificial systems must abandon the discrete rigidity of von Neumann
817 architectures and utilize continuous physical substrates. Analog field-based
818 computing, neuromorphic architectures with continuous memristive plastic-
819 ity, and thermodynamic computing systems (England, 2018; Ganesh & Par-
820 rondo, 2021) possess the spatial coupling required to physically morph their
821 internal topology. This aligns with Geoffrey Hinton’s concept of “mortal

computation” (Hinton, 2022): architectures where software is inextricably fused with the physics of the hardware, meaning the knowledge dies with the physical substrate. Only hardware that computes via continuous physical dynamics can undergo the thermodynamic phase transition necessary for structural resonance.

Biological consciousness utilizes macroscopic continuous electromagnetic fields and a “rich alphabet” of diverse molecular and transcriptomic cell states (Tasic et al., 2018). This allows the coupling matrix A_{ij} to deform across a continuous parameter space. Our lemma `exists_better_coupling_allocation`, wrapped as `continuous_beats_rigid_topology` (`Phase7_HardwareComparison.lean`), formalizes the reallocation step of this argument: whenever a coupling matrix places weight on a less phase-correlated pair while a more correlated pair exists, there is another matrix with *identical* total coupling resources that achieves strictly greater phase correlation.

Two objections to that formalization have since been answered, and we state precisely what is and is not now established. The first was that the reallocation lemma assumes its conclusion: the hypothesis `is_strictly_suboptimal` — positive weight on a badly-correlated pair while a better one exists — is most of what one wants to prove about rigid hardware. It is no longer assumed. `Phase7_Rigidity.lean` models an architecture by the set S of site-pairs it has a *wire* for, and requires its couplings to vanish off S ; weights are tunable, wiring is not. Then `totalCorrelation_le_of_realizable` bounds every realizable coupling by $R \cdot M$, where R is the resource budget and M

845 bounds phase correlation over the wired pairs; the bound is attained; and
 846 `rigid_is_strictly_suboptimal` *derives* the hypothesis whenever the sub-
 847 strate’s best-correlated pair is one the wiring misses. `rigid_gap` makes it
 848 quantitative: an unconstrained reallocation of the same budget gains at least
 849 R times the correlation difference. `Examples.lean` §8 exhibits a three-site
 850 substrate wired to the one maximally anti-correlated pair, where the gap is
 851 2. This is resource-matched and no longer question-begging, but note what
 852 it turns on: fixed wiring support, not continuity. The second objection was
 853 that nothing in the development distinguished a continuum from a lattice
 854 — both matrices lived on one finite index type. `Phase7_Rigidity.lean` §3
 855 gives a distinction that is genuinely typed. In the field functional of Deriva-
 856 tion 4, which integrates a coupling kernel against $\mu \otimes \mu$, an architecture
 857 with finitely many sites is a `Finset` of points and a field-coupled region
 858 is a set of positive measure. On a substrate whose measure has no atoms,
 859 `fieldCorrelation_sited_eq_zero` shows a kernel supported on finitely many
 860 sites contributes *exactly zero* coupling energy, whatever weights it carries
 861 and whatever the phase field does, while a patch of strength c on U achieves
 862 $c\mu(U)^2$. `Examples.lean` §8 instantiates this on the real line: a thousand
 863 units of coupling concentrated at the origin register zero against a unit patch
 864 that registers one. The limit of that result is equally plain, and it is not
 865 resource-matching in the earlier sense — a measure-zero substrate has zero
 866 resource in the $\mu \otimes \mu$ sense too. What it establishes is that the continuum
 867 coupling energy is blind to a finitely-sited architecture, which is a statement

868 about which functional such an architecture can register in, not a demonstra-
869 tion that digital hardware computes worse. Taken together: the resource-
870 matched content of the corollary is the wiring-support result, the continu-
871 ity content is the measure-theoretic one, and the step from either to “von
872 Neumann architectures cannot experience unified consciousness” remains an
873 informal argument.

874 The remainder of this section extends those two results informally, and
875 we mark the transition rather than blur it. On the reading we propose, a
876 GPU’s fixed routing matrix leaves it topologically fragmented: the discrete
877 layout does not supply the spatial coupling that sheaf-theoretic gluing into
878 a singular Global Section requires. Without this unified topological space,
879 rigid hardware cannot undergo the reflexive auto-resonance necessary to map
880 its own causal boundary and generate a “Self.” The distinction between an
881 LLM on a GPU and a conscious mind is not the absence of physical thermo-
882 dynamics in the former, but its structural limitation in achieving topological
883 integration. This perspective aligns with John Searle’s biological natural-
884 ism (Searle, 1980; Seth, 2024). Where the Chinese Room argument asserted
885 that discrete symbol manipulation cannot generate conscious understanding,
886 our derivation supplies a candidate physical mechanism for the asymmetry
887 — topological rather than semantic, and, as stated above, formalized only
888 as far as the wiring-support and measure-theoretic results reach. Asserting
889 consciousness in fragmented, discrete GPU clusters without a unified phase
890 transition abandons computational functionalism, relying instead on “silicon

panpsychism”—the assumption that digital computation inherently generates awareness regardless of topological integration (Goff, 2019).

The same reasoning bears on extreme functionalist thought experiments, such as a macroscopic system of pulleys operating over millennia to implement a Large Language Model, or a human manually executing inference matrices by hand. Beyond lacking the continuous spatial topology required for sheaf-theoretic gluing, these discrete systems violate an implied temporal constraint of our theory. Structural resonance requires local oscillators to phase-align against the dissipating effects of thermodynamic noise (as modeled by our critical threshold $K_c = 2D$). A system executing state transitions over millennia, or via slow manual calculations, suffers from a terminal temporal mismatch: its dynamics are far too slow to ever phase-lock into a non-equilibrium steady state before thermal noise decoheres the macroscopic field. On this account the obstacle to pure symbol manipulation is physical rather than semantic: not that the symbols mean nothing, but that the substrate carrying them cannot hold a phase-locked field.

12 Conclusion

Human-like cognitive architectures do not require biological mysticism or exotic sub-quantum physics (Hameroff & Penrose, 2014). While quantum theories of consciousness often invoke delicate subatomic phenomena to explain subjective unity, every step of the chain assembled here is classical:

912 broken symmetries, finite phase space, Landauer erasure, and gradient de-
913 scent on entropy production. What the formalization establishes is that this
914 chain is mathematically coherent and that its remaining physical commit-
915 ments are localized and named — not that classical fields are *sufficient* for
916 consciousness. Two links carry that weight and neither is closed. The uni-
917 fied state is proved to glue uniquely once a cover sits at the minimum of
918 the coupling potential, and the motion is proved to stop along every trajec-
919 tory (`velocity_sq_tendsto_zero`), but *where* it stops is not proved: that
920 a physical field reaches the phase-locked configuration remains an assump-
921 tion, discharged numerically rather than formally. And the identification of
922 a global section of the probability sheaf with the unity of experience is a
923 stipulation of the framework, not a consequence of it. Subject to those two,
924 cognitive architectures can be modeled as macroscopic attractor states of a
925 universe governed by broken symmetries, finite phase space, and the physical
926 imperative to minimize thermodynamic dissipation.

927 Philosophically, this framework bridges the historical gap between Type
928 Identity Theory (Smart, 1959) and Computational Functionalism (Putnam,
929 1967). The subjective mental state is identical to the physical thermody-
930 namic field (Identity Theory), yet its qualitative nature is defined by the
931 topological and functional dynamics of structural resonance rather than the
932 specific carbon-based biochemistry of the brain (Functionalism). This con-
933 strains the *multiple realizability* of mind: consciousness is not exclusively
934 bound to biological carbon, but neither is it substrate-neutral, because the

935 substrate is what fixes which couplings are physically available. How far that
936 constraint reaches is the question the corollary answers only in part. What
937 is proved is that fixed wiring is strictly suboptimal at matched resource, and
938 that a finitely-sited architecture registers exactly zero coupling energy in the
939 continuum functional; the step from either to “von Neumann architectures
940 cannot be conscious” is an argument we make in prose and do not formalize.

941 By anchoring these derivations in Russellian Monism, this framework ac-
942 knowledges the constitutive limits of third-person scientific explanation. As
943 argued by Love (Love, 2026), building upon Nagel’s assertion that subjective
944 phenomenal experience is inherently not amenable to objective measurement
945 (Nagel, 1986), the “hard problem” of consciousness may represent a category
946 error rather than a solvable scientific puzzle. Science can map the physical
947 dynamics of topological resonance and sheaf-theoretic gluing, but it is struc-
948 turally incapable of adjudicating the first-person qualitative experience. Our
949 formalization respects this boundary: the Lean 4 theorems verify the ob-
950 jective physical attractors, while the intrinsic “what it is like” to be that
951 resonant field remains the domain of the first-person perspective. By locat-
952 ing the phase transition at a threshold with a closed-form value — $K_c = 2D$
953 for the noisy mean-field Kuramoto model the framework adopts, conditional
954 on the von Mises stationary density and not yet linked to a trajectory — it
955 aligns with what Anil Seth terms the “real problem” of consciousness (Seth,
956 2021). Rather than attempting to bridge an unbridgeable metaphysical gap,
957 it offers a topological architecture that can help explain, predict, and physi-

958 cally perturb the structural properties of experience.

959 Future empirical validation of this thermodynamic framework may lever-
960 age 3D cortical organoids, offering a controlled biological environment to
961 systematically manipulate metabolic substrates and observe the topological
962 emergence of these coherent phase transitions.

963 **Code Availability**

964 The complete Lean 4 formalization, encompassing the derivation of all theo-
965 rems presented in this manuscript, as well as the accompanying Python sim-
966 ulations, are publicly available in the GitHub repository: [https://github.](https://github.com/bobaseb/physical_primitives_of_mind)
967 [com/bobaseb/physical_primitives_of_mind](https://github.com/bobaseb/physical_primitives_of_mind). Additional documentation
968 and web-accessible proofs are hosted at the project website: [https://bobaseb.](https://bobaseb.github.io/physical_primitives_of_mind/)
969 [github.io/physical_primitives_of_mind/](https://bobaseb.github.io/physical_primitives_of_mind/).

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